

Trainings ▾

Preparatory Steps

WARNING

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To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the equipment manufacturer service information.
- Drain the oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.

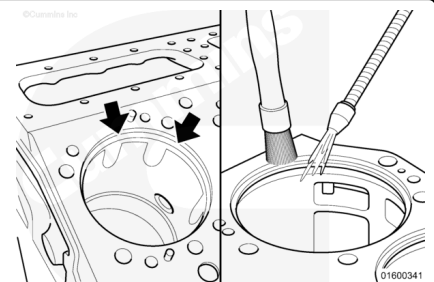
(/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)

- Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Remove the oil suction and transfer tubes. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Remove the oil pump, if necessary. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)
- Remove the block stiffener plate.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a scraper or a fibrous abrasive pad, such as Scotch-Brite™ 7477, or equivalent, and solvent. Clean the top deck of the block. Do **not** allow any dirt into the cylinder. The surface **must** be free of dirt, oil, and gasket material, but does **not** have to look like new metal.

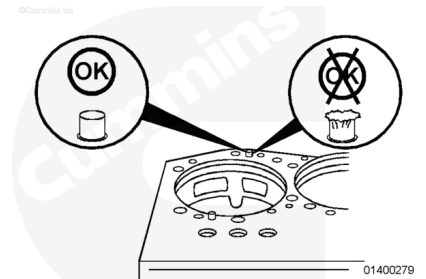
Note : The counterbore ledge **must** be clean and dry. The sealant used in assembly will **not** dry if oil is present.

Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired. See the Alternative Repair Manual, Bulletin 3379035.

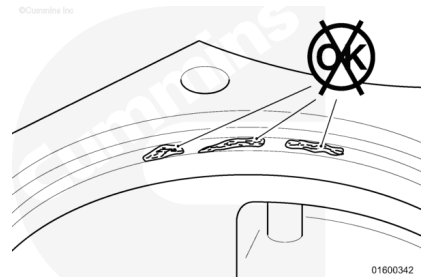
Fretting damage in any other area is acceptable **only** if it does **not** change the protrusion measurement of the counterbore or liner.

Inspect the dowel pins for damage.

Use the dowel pin extractor, Part Number ST-1134, or equivalent, to remove the pins if they are damaged.

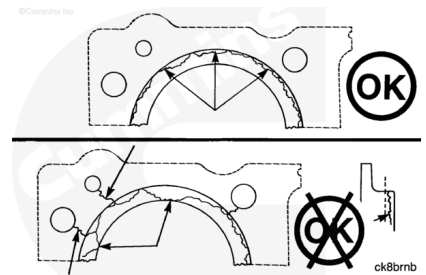


Inspect the counterbore ledge for pitted areas. The ledge **must** be machined if pitting is present.

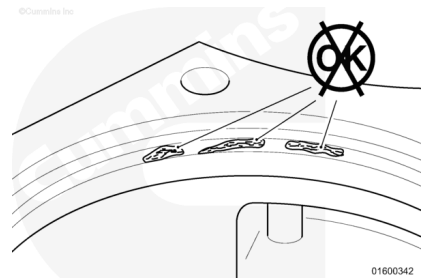


Check for cracks.

Use a Crack Detection Kit, Part Number 337532, or equivalent, and check the counter bore ledges for cracks. Circumferential cracks are acceptable if they do **not** extend more than 1/2 of the distance across the ledge.



Cracks that extend into a capscrew hole are **not** acceptable. The block can be used if a capscrew hole thread insert is installed in the capscrew hole. See the Alternative Repair Manual, Bulletin 3379035.

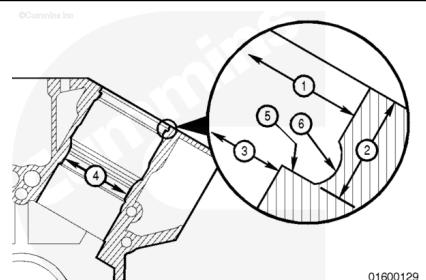


Inspect the counterbore

To inspect the block properly, the counterbore design **must** be determined.

Terms used are:

1. Upper counterbore inside diameter
2. Counterbore depth
3. Lower counterbore inside diameter
4. Packing ring bore
5. Counterbore ledge
6. Counterbore radius.

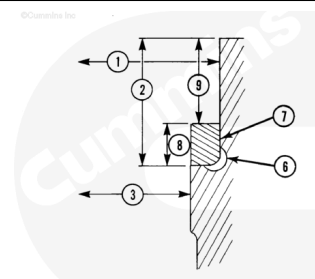


Measure

Counterbore Ring - Thick Deck Block Design, **most** K38 engines with engine serial number first greater than 33100193 contain this design.

All K50 engines contain this design.

(1) The diameter depends on whether standard size or oversize liners were installed.



ck6brsb

Oversize			
	mm		in
Standard	190.28	MIN	7.492
	190.34	MAX	7.494
0.010	190.53	MIN	7.502
	190.59	MAX	7.504
0.020	190.79	MIN	7.512
	190.85	MAX	7.514
0.030	191.04	MIN	7.522
	191.10	MAX	7.524
0.040	191.30	MIN	7.532
	191.36	MAX	7.534

(2) The actual size depends on whether standard size or oversize liners were installed. The size also depends on whether the block was cut for shims

Oversize [Inch]			
	mm		in
Standard	18.26	MIN	0.718
	18.31	MAX	0.720
0.010	18.52	MIN	0.728
	18.57	MAX	0.730
0.020	18.77	MIN	0.738
	18.82	MAX	0.740
0.030	19.02	MIN	0.748
	19.07	MAX	0.750
0.040	19.28	MIN	0.758

	19.33	MAX	0.760
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The depth can be a maximum of 3.39 mm [0.133 inch] greater if the block was machined previously for shims and oversize liners.

(3) Lower Counterbore Inside Diameter		
mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028.

Lower Counterbore Inside Diameter		
mm		in
181.80	MIN	7.157
181.81	MAX	7.158

Note : The cylinder block can be used again only if the correct press fit between the cylinder liner and dimensions (1) is obtained.

(6) This design contains a double undercut radius.

(7) This design contains a counterbore ring that functions as a load distribution washer.

(8) Counterbore ring height.

Oversize [Inch]			
	mm		in
Standard	5.055	MIN	0.1990
	5.067	MAX	0.1995
0.002	5.105	MIN	0.2010
	5.118	MAX	0.2015
0.004	5.156	MIN	0.2030
	5.169	MAX	0.2035

(9) This depth depends on the size of the liner. Liner protrusion **must** be 0.152 to 0.203 mm [0.006 to 0.008 in].

Oversize [Inch]			
	mm		in
Standard	13.16	MIN	0.519
	13.23	MAX	0.521

0.010	13.41	MIN	0.529
	13.49	MAX	0.531
0.020	13.67	MIN	0.539
	13.74	MAX	0.541
0.030	13.92	MIN	0.549
	14.00	MAX	0.551
0.040	14.17	MIN	0.559
	14.25	MAX	0.561

If the upper counterbore is **not** within standard specifications, the block **must** be machined for oversize liners.

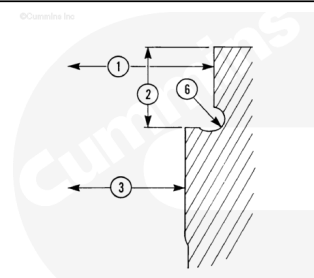
If the block is larger than the specification for the largest oversize, a repair sleeve can be installed. Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims. Before machining the block for shims, check to see if the protrusion can be corrected by installing an oversize counterbore ring.

⚠ CAUTION ⚠

Do not attempt to machine blocks of this design for a counterbore ring. It will cause the block to break.



ck4brgb

Thin Deck Block - Double Undercut Radius Design, some K38 standby generator set engines with an engine serial number first higher than 33100193 contain this design.

No K50 engines contain this design.

(1) The actual size depends on whether the standard size or oversize liners were installed.

Oversize			
	mm		in
Standard	190.277	MIN	7.4912
	190.343	MAX	7.4938
0.010	190.597	MIN	7.5038
	190.663	MAX	7.5064
0.020	190.917	MIN	7.5164
	190.983	MAX	7.5190

0.030	191.237	MIN	7.5290
	191.303	MAX	7.5316
0.040	191.557	MIN	7.5416
	191.623	MAX	7.5442

(2) These specifications are required for a liner protrusion of 0.152 to 0.203 mm [0.006 to 0.008 inch]. The protrusion specification changed since blocks this design were introduced.

The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.

Oversize [Inch]			
	mm		in
Standard	13.16	MIN	0.519
	13.23	MAX	0.521
0.010	13.41	MIN	0.529
	13.49	MAX	0.531
0.020	13.67	MIN	0.539
	13.74	MAX	0.541
0.030	13.92	MIN	0.549
	14.00	MAX	0.551
0.040	14.17	MIN	0.559
	14.25	MAX	0.561

(3) Lower Counterbore Inside Diameter

mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028.

Lower Counterbore Inside Diameter		
mm		in
181.80	MIN	7.157
181.81	MAX	7.158

(6) This design contains a double undercut radius.

If the upper counterbore is **not** with standard specifications, the block **must** be machined for shims.

If **only** the counterbore depth (2) is **not** within specifications, the block **must** be machined for shims.

Refer to Procedure 001-026 in Section 1
(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html) for Cut Counterbore Depth for shims.

If the block is larger than the specification for the largest oversize liner, a repair sleeve can be installed.

Refer to Procedure 001-058 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)

It is recommended that a block of this design be used **only** for K38 standby generator set applications.

⚠ CAUTION ⚠

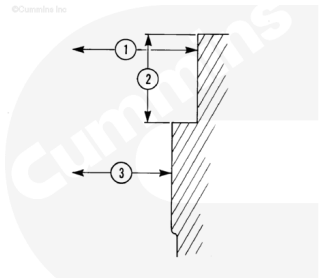
Do not attempt to machine blocks of this design for a counterbore ring. It will cause the block to break.

Thin Deck Block Design, **most** K38 engines with an engine serial number first lower than 33100193 contain this design.

No K50 engines contain this design.

(1) The actual size depends on whether the standard size or oversize liners were installed.

Oversize [Inch]			
	mm		in
Standard	190.284	MIN	7.4915
	190.335	MAX	7.4935
0.010	190.538	MIN	7.5015
	190.589	MAX	7.5035
0.020	190.792	MIN	7.5115
	190.843	MAX	7.5135
0.030	191.046	MIN	7.5215
	191.097	MAX	7.5235
0.040	191.300	MIN	7.5315
	191.351	MAX	7.5335



Note : These specifications are those required for a liner protrusion of 0.152 to 0.203 mm [0.006 to 0.008 inch]. The protrusion specification has changed since blocks of this design were produced.

(2) The actual size depends on whether the standard size or the oversize liners were installed, and if the block was machined for shims.

Oversize [Inch]			
	mm		in
Standard	13.16	MIN	0.519
	13.23	MAX	0.521
0.010	13.41	MIN	0.529
	13.49	MAX	0.531
0.020	13.67	MIN	0.539
	13.74	MAX	0.541
0.030	13.92	MIN	0.549
	14.00	MAX	0.551
0.040	14.17	MIN	0.559
	14.25	MAX	0.561

(3) Lower counterbore inside diameter

mm		in
181.74	MIN	7.155
181.79	MAX	7.157

For engines with engine serial number lower than 33101028.

Lower counterbore inside diameter		
mm		in
181.80	MIN	7.157
181.81	MAX	7.158

Note : The cylinder block can be used again **only** if the correct press fit between the cylinder liner and dimensions (1) is obtained.

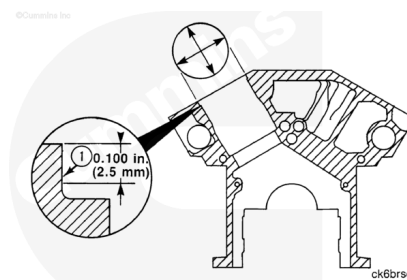
This design does **not** contain the double undercut radius.

It is recommended that a block of this design be used **only** for K38 standby generator set applications or those applications requiring a rated power of 675 kW [900 bhp] or less.

Measure the inside diameter of the upper counterbore (1). The point of measurement **must** be within 2.5 mm [0.100 inch] from the top surface of the block.

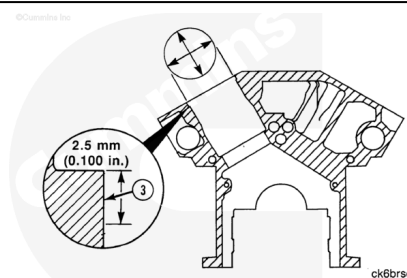
The diameter of the upper counterbore **must** be round within 0.025 mm [0.001 inch]. If the measurement is **not** within specification, determine if the block can be machined for oversize liners. See Procedure cylinder block - Cut for Oversize Liners.

Note : The upper counterbore **must** be no more than 0.025 mm [0.001 inch] larger than the cylinder liner flange, or no more than 0.076 mm [0.003 inch] smaller than the cylinder liner flange.

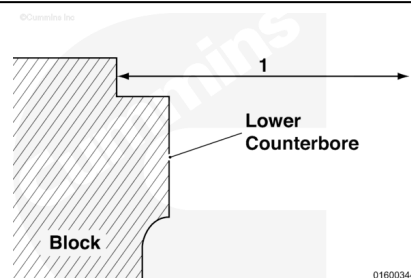


Measure the inside diameter of the lower counterbore (3). The point of measurement **must** be within 2.5 mm [0.100 inch] from the top of the counterbore ledge. Measure at two locations, 90 degrees apart

The diameter of the lower counterbore **must** be within 0.025 mm [0.001 inch].



If the diameter of the lower counterbore is larger than the specification, the block can be used **only** if the outside diameter of the liner flange is 0.025 mm [0.001 inch] larger than the inside diameter of the upper counterbore. If there is **not** a press fit between the liner and the upper counterbore, the block **must** be machined for oversize liners (dimension 1).



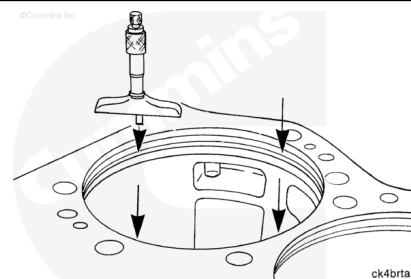
⚠ CAUTION ⚠

Make sure the micrometer contacts the flat surface of the counterbore ledge. It must not touch the counterbore face.

Use a depth micrometer. Measure the counterbore depth at the four places shown. The four measurements **must not** vary more than 0.025 mm [0.001 inch]. If the measurements are **not** in accordance with the specifications, the counterbore ledge **must** be machined.

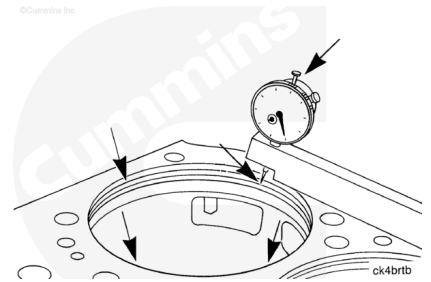
Refer to Procedure 001-058 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html)



⚠ CAUTION ⚠

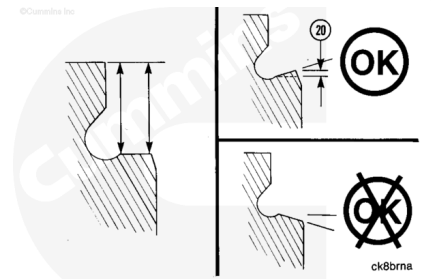
Make sure the indicator does not contact the counterbore face (radius) on a block that does not have a double undercut.



Use a Dial Depth Gauge, Part Number ST-537. Measure the angle of the counterbore ledge at no less than four locations around the ledge. One measurement at each location **must** be as close as possible to the counterbore edge.

It is acceptable if the counterbore ledge has an angle no greater than described as follows:

- If the distance measured near the counterbore radius (face) is the same or no more than 0.036 mm [0.0014 inch] farther than the distance measured near the counterbore ledge, the counterbore ledge is acceptable.
- The counterbore ledge angle is **not** acceptable if the distance near the counterbore edge is farther than the distance measured near the counterbore radius (face). In this case, the ledge **must** be machined. See the Alternative Repair Manual, Bulletin 3379035 for repairing the packing ring bore with a bushing.

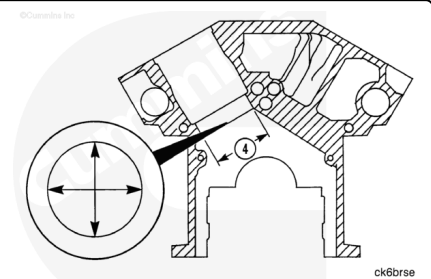


Measure the packing ring bore.

Closely inspect the chamfer at the top of the packing ring bore.

Excessive pitting of the chamfer or the ring bore is **not** acceptable. See the Alternative Repair Manual, Bulletin 3379035, for repairing the packing ring bore with a bushing.

Measure the diameter of the packing ring bore.



Packing Ring Bore diameter

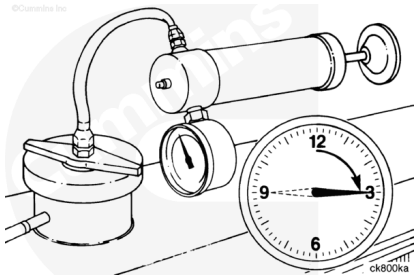
mm		in
177.32	MIN	6.981
177.48	MAX	6.987

Leak Test

Pressurize the engine cooling system.

Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)

Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Make sure the system is holding air pressure before beginning inspection.



Measurements

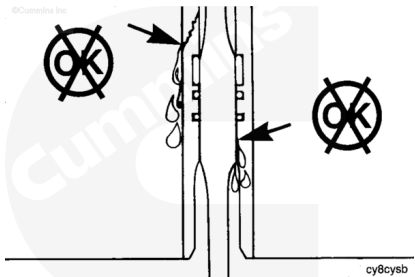
	kpa	psi
Air Pressure	138	20

Inspect the outside diameter of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a leak is found, remove the cylinder liner(s). Inspect the sealing ring and cylinder liner.

Refer to Procedure 001-028 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-028-tr.html)

Inspect the cylinder block liner counterbore area.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the block stiffener plate.

- Install the oil pump, if removed. Refer to Procedure 007-031 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-031-tr.html)
- Install the oil suction and transfer tubes. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Fill the engine with clean engine oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html)
- Connect the batteries. Refer to the equipment manufacturer service information.
- Operate the engine until it reaches a temperature of 80°C [176°F].
- Check for coolant and lubricating oil leaks.